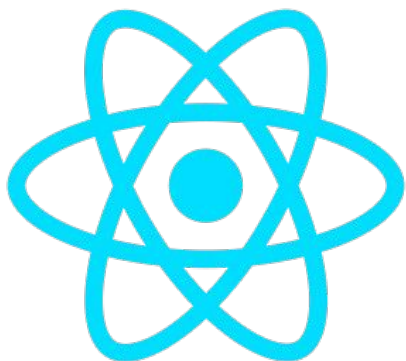




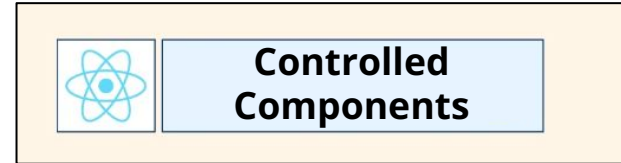
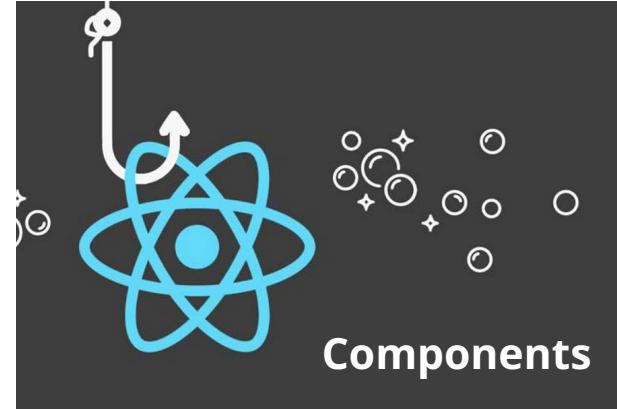
The Ultimate React Course



Lecture 19: Controlled Component

Table of Contents

- Quick Recap
- Who controls Input?
- Controlled Input
- Common Bug
- Forms In React
- HTML form problem
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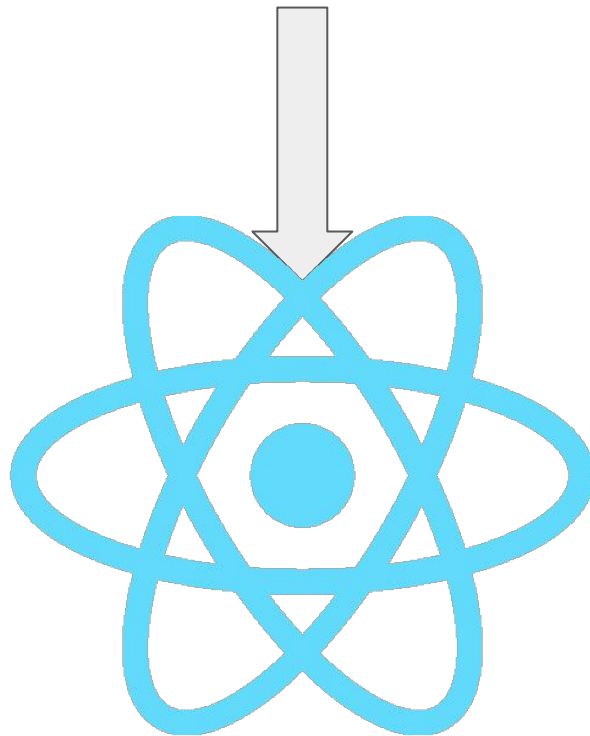


Who Controls your input?



Newton School
of Technology

Browser 🤔 or React 



Class Question

When you type in Google Search Where Does Your Value Lives?

- Inside the **input**?
- In the **Browser**?
- **Somewhere in the app**?



Reality Check

The value does not live only inside the input.....!



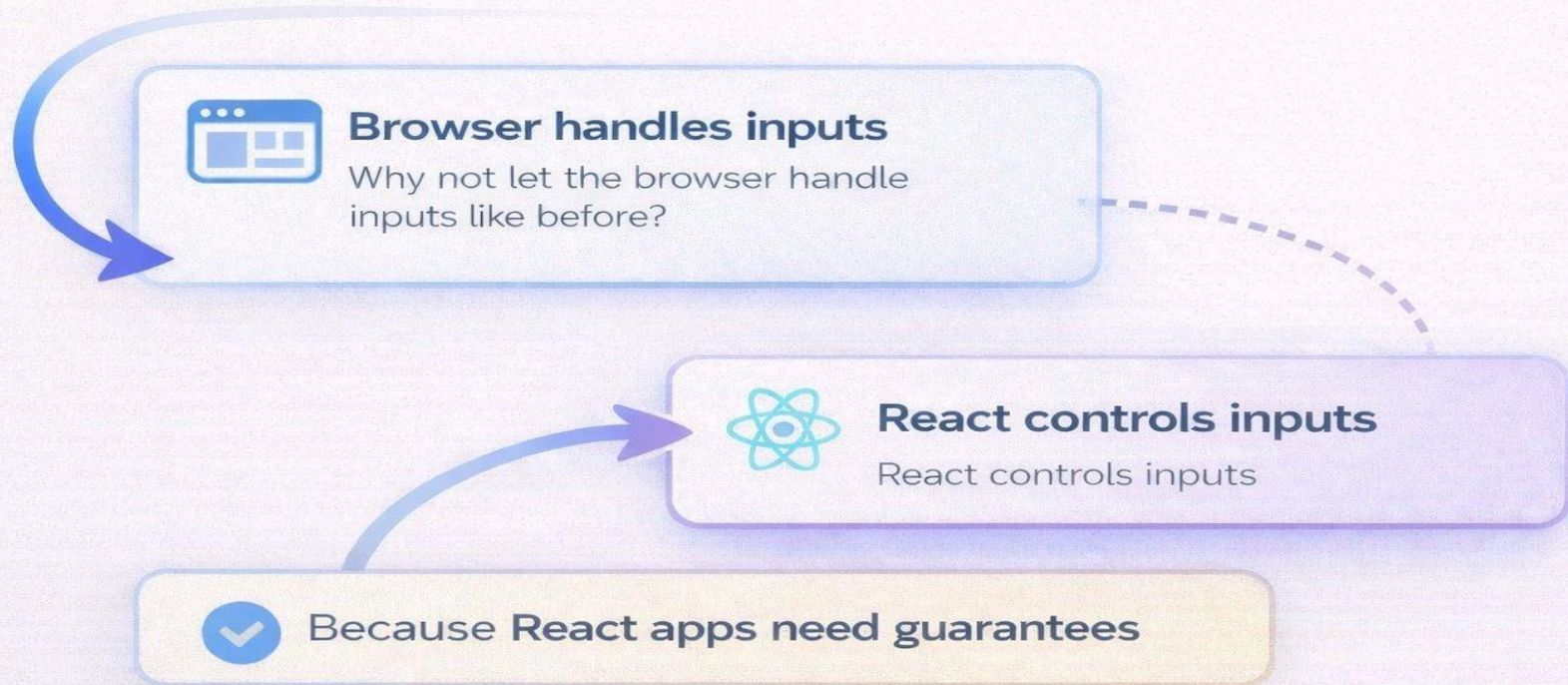
Key Truth



In React...

State controls the UI — not the browser

Why control matters ?



What React Gains

When React owns input value, it can:

- Validate while typing
- Show instant errors
- Enable / disable buttons
- Build live search
- Power login & signup forms

These are app features, not
browser features



Controlled Input



- ❖ An input whose value comes from React state

Visual Flow

value = state

UI $\leftarrow$ state $\rightarrow$ user

Browser is no longer the boss



VALUE



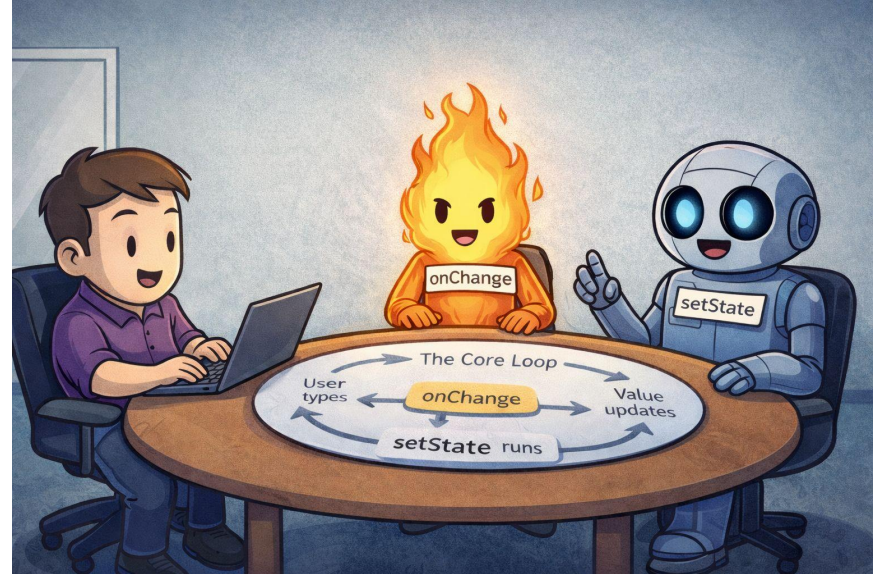
UI



USER

The Core Loop

- ★ User types
 - **onChange** fires
 - **setState** runs
 - **value** updates
 - **UI re-renders**
- ★ This loop is the heart of React forms



Giving React Vision



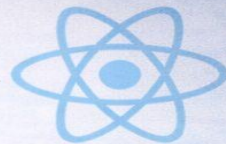
Step 1: Create state

```
const [name, setName] = useState('');
```

- ❑ **React now has memory
But input isn't
connected yet**

Input Freezes

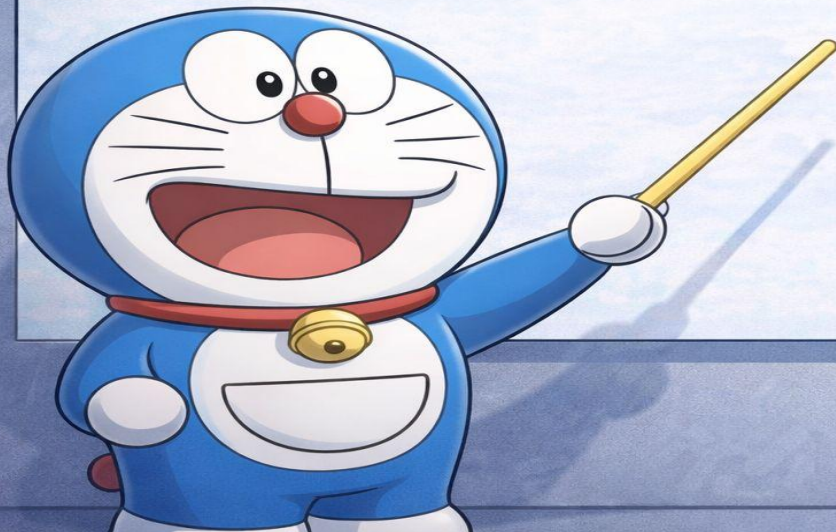
Bind value only 📌



```
<input value={name} />
```

😬 Typing stops

- Why?
React controls value
But update logic is
missing (function is
missing)



Completing the Loop

php-template

```
<input  
  value={name}  
  onChange={(e) => setName(e.target.value)}  
>
```

- Now it works!
- Input shows what React tells it to show

Time to test Yourself

Do the following

- Change placeholder
- Set default name
- Display Output



Common Bug



My input stopped typing!

✗ Missing onChange

✓ Controlled input must update state via function

Quick Check

Which prop makes an input controlled?

```
<input value={state} onChange={...} />
```

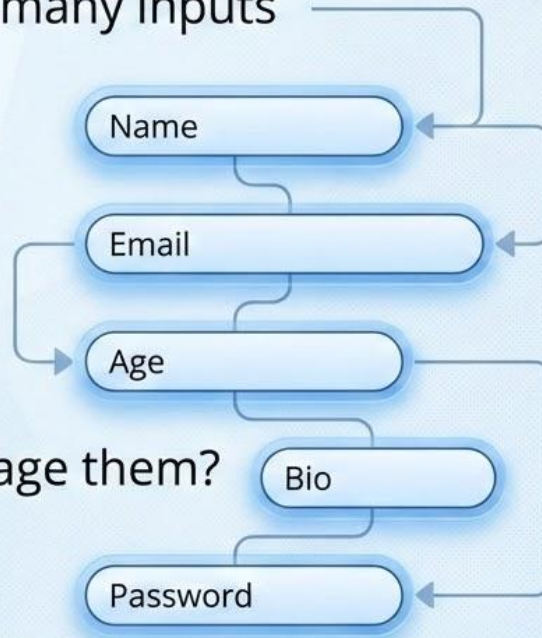
Remainder:

Value+onChange

Real Forms

Real forms have many inputs

- Name
- Email
- Age
- Bio
- Password



How do we manage them?



How do we manage them?

One State per Field

```
const [name, setName] = useState("");  
const [email, setEmail] = useState("");  
const [age, setAge] = useState("");
```

✓ Easy to understand

✗ More variables



One Object State

```
const [form, setForm] = useState({  
  name: "",  
  email: "",  
  age: ""  
});
```



⚠ Needs careful updates

⚠ Easy to overwrite data

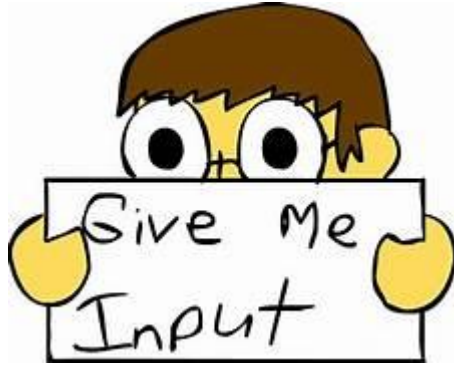
Correct Object Update

SCSS

```
setForm({  
  ...form,  
  age: e.target.value  
});
```

Always spread old state

Student Task (Multi-Input)



Build a form with:

- Age (number)
- Email
- Bio (textarea)

Rules:

All inputs must be controlled

Debugging Story

Only one field updates!

- ✗ Old object overwritten
- ✓ Spread previous state



Different Input Types



Forms also include:

- Checkbox
- Radio
- Select dropdown

Each behaves differently

Controlled Checkbox

Checkbox uses
checked, not value

php-template

```
<input  
  type="checkbox"  
  checked={isChecked}  
  onChange={(e) => setIsChecked(e.target.checked)}
```

Radio Buttons

pgsql

```
<input  
  type="radio"  
  value="Student"  
  checked={role === "Student"}  
>
```

✓ Compare value with state

Controlled Select

php-template

```
<select value={role} onChange={...}>  
  <option>Student</option>  
  <option>Teacher</option>  
</select>
```

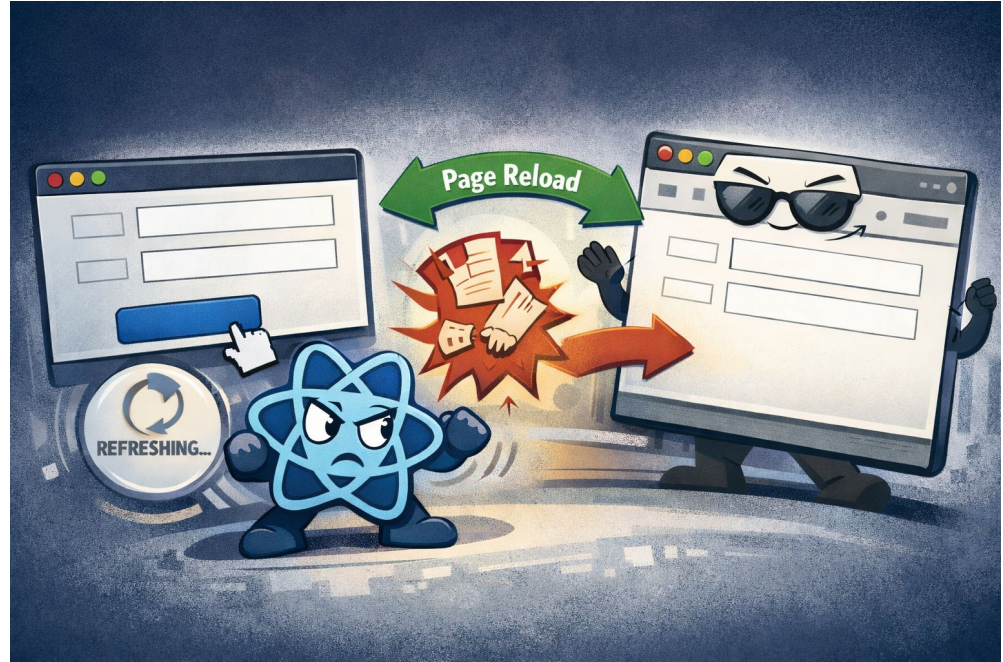


HTML Form Problem

On submit:

- Page refreshes
- State is lost
- Browser takes control

✗ Not what React wants



Preventdefault

```
function handleSubmit(e) {  
  e.preventDefault();  
}
```



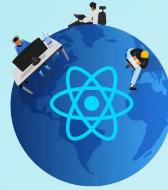
React says:
“Don’t refresh.
I’ve got this.”

React Form Submission

```
<form onSubmit={handleSubmit}>
```



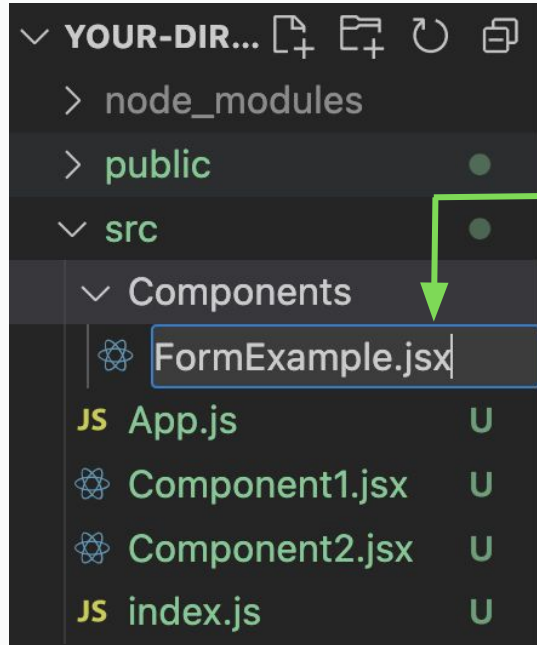
All Data already in state
Just process it



Let's Get Coding

Forms: Creating the file

Forms in React are different from regular HTML because React **manages form elements using state**. Let's create a *FormExample.jsx* component and include it in our App.js



Go to Explorer and create a folder Components and inside it create a component named FormExample.jsx

Include a form in that file.

```

1  function FormExample() {
2
3      return (
4          <form>
5              <label>
6                  Name:
7                  <input type="text" placeholder="Name" />
8              </label>
9              <button type="submit">Submit</button>
10         </form>
11     );
12 }
13
14 export default FormExample;
  
```

Forms: Creating and including in App.js

Let's add FormExample in our App.js as a component:

```

1  function FormExample() {
2
3    return (
4      <form>
5        <label>
6          Name:
7          <input type="text" placeholder="Name" />
8        </label>
9        <button type="submit">Submit</button>
10     </form>
11   );
12 }
13
14 export default FormExample;

```



```

1  import FormExample from "../Components/FormExample";
2
3  function App() {
4    return (
5      <div>
6        <FormExample />
7      </div>
8    );
9  }
10
11  export default App;

```

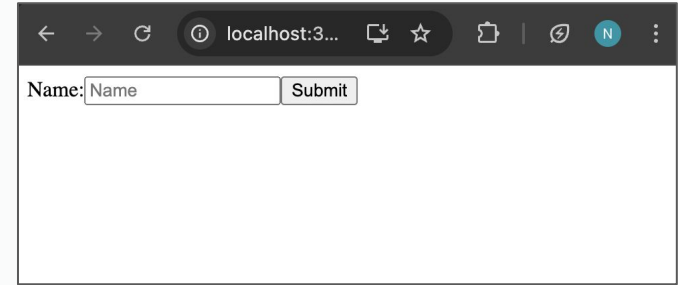
Importing FormExample
using a **Relative path**

Including
FormExample in
App.js

Forms: Viewing in browser

Let's have a look at it in the browser:-

```
1 import FormExample from "../Components/FormExample";
2
3 function App() {
4   return (
5     <div>
6       <FormExample />
7     </div>
8   );
9 }
10
11 export default App;
```



The browser window shows a form with a text input field labeled "Name" and a "Submit" button. The browser's address bar displays "localhost:3...".

Forms in React

Create a state variable name and its setter function setName. And add it to the value attribute to our input box.

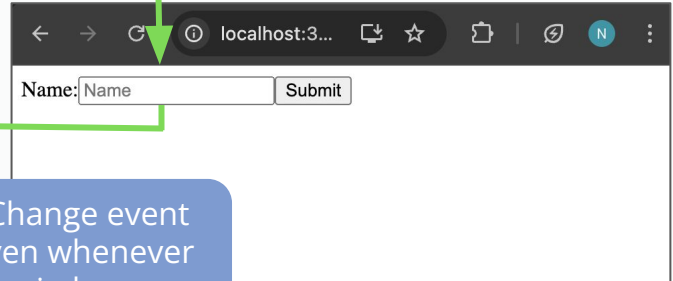
```

1  function FormExample() {
2    const [name, setName] = useState("");
3
4    return (
5      <form>
6        <label>
7          Name:
8          <input
9            type="text"
10           value={name}
11           placeholder="Name"
12         />
13        </label>
14        <button type="submit">Submit</button>
15      </form>
16    );
17  }
18
19  export default FormExample;

```

When name state variable changes, value in the input box also changes too

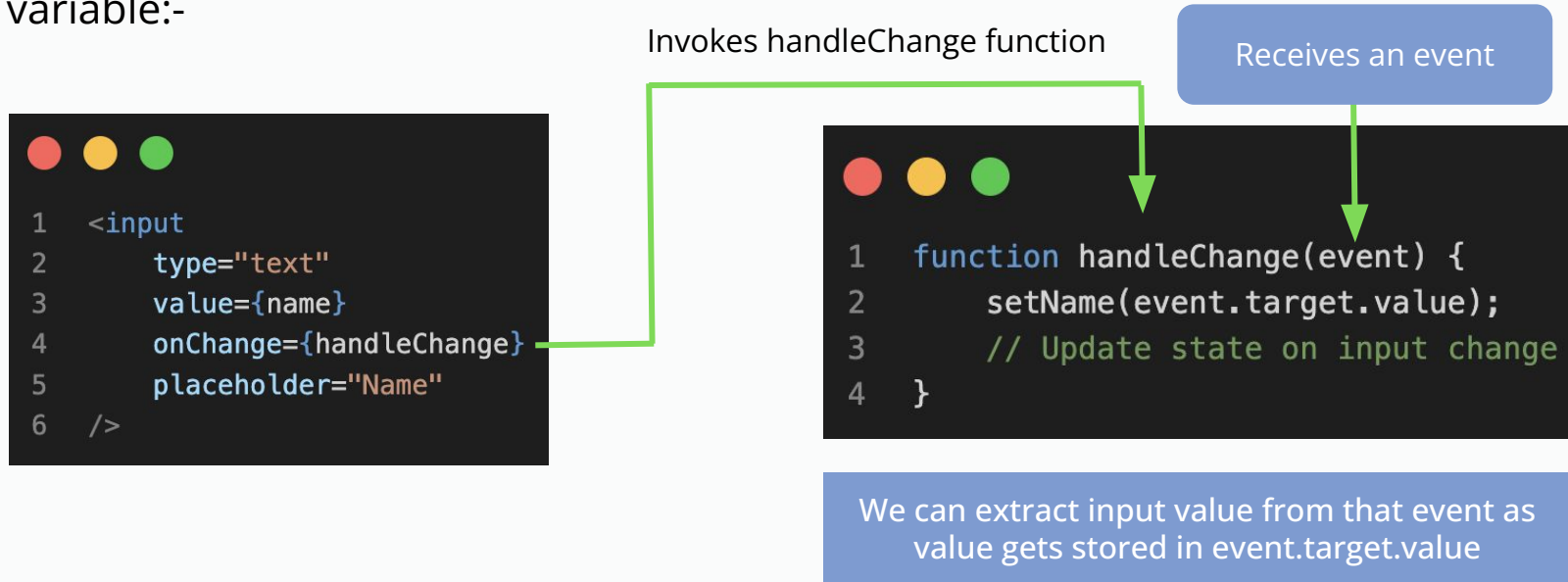
Now we need to change the name state variable whenever user types a value in input box



We can add an onChange event which throws an even whenever we change value in box

Forms in React

Add onChange event and associate a function which changes the value of name state variable:-



Forms in React

Let's also add an event on form submission:-

```
function FormExample() {  
  const [name, setName] = useState("");  
  
  const handleSubmit = (event) => {  
    event.preventDefault();  
    console.log('Submitted:', name );  
  };  
  
  function handleChange(event) {  
    setName(event.target.value);  
    // Update state on input change  
  }  
  
  return (  
    <form onSubmit={handleSubmit}>  
      <label>  
        Name:  
        <input type="text" value={name} onChange={handleChange} />  
      </label>  
      <button type="submit">Submit</button>  
    </form>  
  );  
}
```

We need to prevent the default behaviour of submit using `event.preventDefault()`

When we submit the form `handleSubmit` function is invoked and by default an event is passed to it.

Final Takeaways

- ✓ What controlled components are
- ✓ Why React must own values
- ✓ How onChange + state works
- ✓ Multi-input form patterns
- ✓ How to debug form issues

If React doesn't own the value...
It can't guarantee correctness.



In Class Questions

**Thanks
for
watching!**

Feedback

Now it's time to give your valuable feedback

